

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mark  | Comments                                                                                                                                                                                              |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1      | 1. <u>Polysaccharide</u> of <u><math>\alpha</math>-glucose</u> ;<br><b>OR</b><br><u>polymer</u> of <u><math>\alpha</math>-glucose</u> ;<br>2. (Joined by) glycosidic bonds<br><b>OR</b><br>Branched structure;                                                                                                                                                                                                                                                                                      | 2     |                                                                                                                                                                                                       |
| 2.2      | 1. Hydrolysed (to glucose);<br>2. Glucose used in respiration;                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     | 1. Ignore 'Broken down'<br>2. 'Energy produced' disqualifies mp2                                                                                                                                      |
| 2.3      | 1. Membrane folded <b>so</b> increased/large surface area;<br><b>OR</b><br>Membrane has increased/large surface area <b>for</b> (fast) diffusion/facilitated diffusion/active transport/co-transport;<br>2. Large number of <u>protein</u> channels/carriers (in membrane) <b>for</b> facilitated diffusion;<br>3. Large number of <u>protein</u> carriers (in membrane) <b>for</b> active transport;<br>4. Large number of <u>protein</u> (channels/carriers in membrane) <b>for</b> co-transport; | 2 max | 1. Accept 'microvilli to increase surface area'<br>1. Reject reference to villi.<br>Note feature and function required for each marking point and reference to large/many/more.<br>List rule applies. |
| 2.4      | $3.3 \times 10^{-5}$ <b>OR</b> $3.28 \times 10^{-5}$ <b>OR</b> $3.281 \times 10^{-5}$ ;;<br>1 mark for<br>Evidence of 128 (cells)<br>Correct numerical calculation but not in standard form gains 1 mark ( $0.00003281$ <b>OR</b> $0.0000328$ <b>OR</b> $0.000033$ );                                                                                                                                                                                                                               | 2     | Accept any number of significant figures as long as rounding correct ( $3.28125 \times 10^{-5}$ scores 2 marks)                                                                                       |

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